Sharan Teja Medikar

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PROFESSIONAL SUMMARY

AI Engineer building evaluated LLM and agentic systems with a growing focus on reliability-first AI infrastructure. Improved Qwen2.5-Coder on sanitised MBPP from 71.9% single-pass to 82.44% with adaptive execution-feedback refinement; currently building ReliableOps Agent with PostgreSQL/Redis/Celery durability patterns validated by 195 passing tests, alongside operational software used across approximately 30 patrol vehicles and 60+ users.

TECHNICAL SKILLS

AI / ML: Large Language Models, Generative AI, AI Agents, Machine Learning, Deep Learning, Computer Vision, Vision-Language Models, Model Evaluation, Benchmarking, Prompt Engineering, Fine-Tuning / Model Adaptation
AI Systems / Backend: Python, PostgreSQL, SQLAlchemy, Redis, Celery, Docker, Alembic, Pytest, Playwright, SQLite, APIs, Automated Testing
Frameworks / Data: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, SQL, Git, Linux, Bash, JavaScript, TypeScript

EDUCATION

MSc Artificial Intelligence | University of Surrey | Sep 2025 - Sep 2026

BTech Computer Science and Engineering | National Institute of Technology Silchar | Jun 2021 - May 2025

PROJECTS

ReliableOps Agent - Reliability-First Agentic AI Operations System

In Development | Python, PostgreSQL, SQLAlchemy, Redis, Celery, Docker, Alembic, Pytest

- Developing an AI-enabled operations platform for SaaS billing and subscription exceptions, with durable workflow execution, tenant isolation and reliable asynchronous processing.
- Implemented a PostgreSQL transactional outbox and durable consumer inbox with idempotent consumption, lease-based event claiming, retry handling and crash recovery for Redis/Celery task processing.
- Built workflow-dispatch intent persistence and tenant-scoped retrieval with deterministic ordering and pagination, backed by transaction-safety and tenant-isolation tests.
- Validated duplicate delivery, rollback behaviour, PostgreSQL interruption and worker SIGKILL recovery through unit, integration and fault-injection tests; 195 tests passed on two consecutive isolated-database runs.
- Ran Pyright and focused Ruff checks successfully on the new implementation, complementing the repeatable isolated-database test runs.

Self-Refining Code Generation Using Execution Feedback

MSc Artificial Intelligence Dissertation | University of Surrey

- Built an execution-feedback code-generation system in Python and PyTorch that generated, executed and iteratively repaired candidate programs using Qwen2.5-Coder-7B-Instruct and CodeLlama-13B-Instruct.
- Compared single-pass generation, compute-matched Best-of-5, adaptive refinement and fixed-k refinement across HumanEval, MBPP, HumanEval Pro and MBPP Pro using execution-derived feedback and deterministic stopping logic.
- Improved Qwen2.5-Coder on sanitised MBPP from 71.9% single-pass to 82.44% with adaptive execution-feedback refinement, and HumanEval Pro from 65.24% to 70.12%; analysed failure modes, regressions and convergence to identify when refinement helped or degraded accuracy.
- For CodeLlama on MBPP, tracked 257/427 problems ever solved versus 211/427 final-only, exposing 46 regressions and showing why final-score evaluation alone can hide unstable refinement behaviour.

AI Job Application Agent

Personal / Applied AI Engineering Project | Python, Playwright, SQLite

- Built an AI-assisted job-search workflow combining vacancy discovery, deterministic eligibility filtering, semantic matching/scoring, evidence-grounded application generation and Playwright-based browser interaction with persistent SQLite state.
- Implemented duplicate prevention, structured-output/schema validation, automated tests, bounded retries/deadlines, deterministic safeguards and human approval boundaries for ambiguous or consequential actions.
- Separated deterministic eligibility and safety logic from model judgement while preserving source provenance and auditable workflow state transitions.

PROJECTS

Vision-Language Model Prompt Learning & Adaptation

CLIP | Prompt / Parameter-Efficient Adaptation | Model Evaluation

- Implemented and compared prompt and parameter-efficient adaptation methods including CoOp, CoCoOp, KgCoOp, CuPL, PromptSRC, ProGrad, LoRA, Tip-Adapter, CLIP-Adapter and MaPLe across six image-classification datasets.
- Improved EuroSAT from 53.99% zero-shot to 89.70% with model adaptation and achieved 91.23% Top-1 accuracy on Oxford Flowers with CoOp.
- Compared transfer behaviour across EuroSAT, Oxford Flowers, Food101, DTD, Stanford Cars and FGVC Aircraft to assess prompt learning and parameter-efficient adaptation across visual domains.

Physics-Informed Environmental Machine Learning

AI & Sustainability | Applied Machine Learning

- Built an environmental ML comparison across 28,632 site-year observations spanning 26,446 locations, evaluating XGBoost, LSTM and physics-informed LSTM approaches using a year-based 2023/2024 split.
- Conducted interpretability and environmental-equity analysis across deprivation groups, examining model behaviour and subgroup error patterns alongside the modelling experiment.
- Reduced model size from approximately 1.8 MB to 908 KB using float16; review of the experiment highlighted the importance of enforcing the prediction boundary in both temporal splitting and feature construction.

AI & Health - ELSA Mental-Health Prediction

Python | scikit-learn | Supervised Machine Learning

- Built and evaluated supervised ML models on 10,838 ELSA participants using 5-fold cross-validation; Gradient Boosting achieved the best verified AUC of 0.853.
- Evaluated a Wave 1 to Wave 3 longitudinal cohort containing 6,511 participants, keeping cohort definition and predictive-modelling provenance explicit.

EXPERIENCE

Founder & Technical Lead | Police Night Patrolling System

Government of Assam stakeholder project | Dec 2023 - Sep 2024

- Originated and pitched a real-time police night-patrolling platform, secured district-government stakeholder approval and assembled a five-person engineering team.
- Worked directly with government stakeholders through development, testing, user acceptance and technical handover, contributing to authentication, backend and real-time functionality using Flutter, Firebase and Express.js.
- System reached operational use across approximately 30 patrol vehicles and 60+ users.

Software Engineering Intern | Fleckor Tech

Hyderabad, India | Jun 2024 - Jul 2024

- Contributed frontend and backend components across 3+ core product features and API integrations within an existing development environment.
- Resolved approximately 15-20 software defects within sprint timelines and supported application testing/deployment preparation.

Freelance Software Developer

2022 - 2025

- Delivered approximately 25-30 websites and 10-12 applications for clients and small businesses, including complete builds and modifications to existing software.
- Worked across frontend, backend, full-stack and UI/UX development based on client requirements.